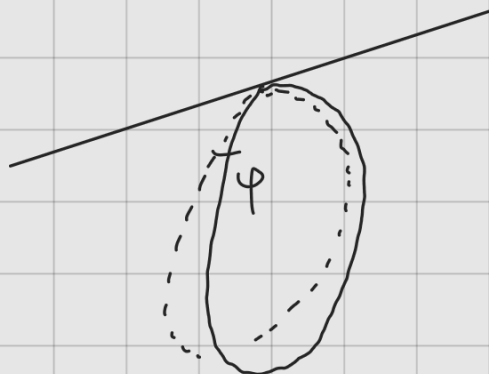


N69

Найти: $\frac{T_1}{T_2}$ | Решение:

1)



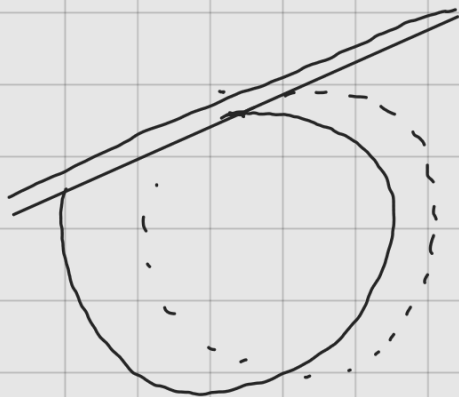
$$\Delta U = mgR(1 - \cos \varphi) = mgR \frac{\varphi^2}{2} = \frac{mgR\varphi^2}{2}$$

$$\Delta K = \left(m \frac{R^2}{2} + mR^2 \right) \dot{\varphi}^2 = \frac{3mR^2}{2} \dot{\varphi}^2$$

$$\omega_1 = \sqrt{\frac{mgR}{3mR^2}} = \sqrt{\frac{2 \cdot g}{3 \cdot R}} = \sqrt{\frac{2}{3} \frac{g}{R}} \Rightarrow$$

$$\Rightarrow T_1 = 2\pi \sqrt{\frac{3R}{2g}}$$

2)



$$\Delta U = \frac{mgR}{2} \varphi^2$$

$$\Delta K = \frac{(mR^2 + mR^2) \dot{\varphi}^2}{2} = \frac{2mR^2 \dot{\varphi}^2}{2}$$

$$\omega_2 = \sqrt{\frac{mgR}{2mR^2}} = \sqrt{\frac{g}{2R}}$$

$$3) T_1 / T_2 = \sqrt{\frac{1}{3} \cdot 4} = \frac{\sqrt{3}}{2}$$

$$\sqrt{\frac{3R}{2g}} \cdot \sqrt{\frac{g}{2R}}$$

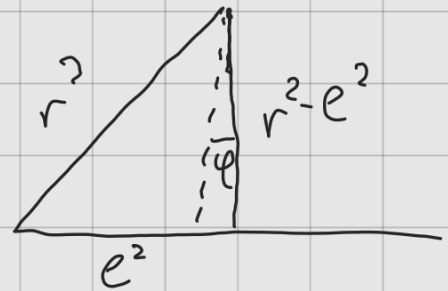
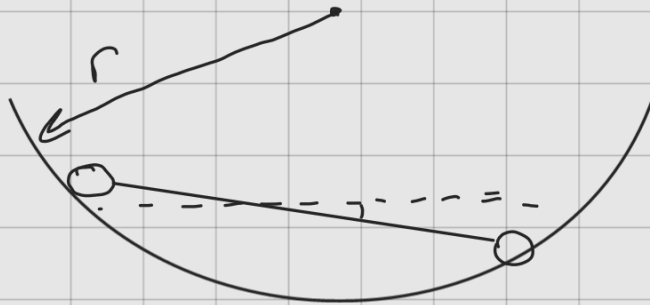
N70

Дано: | Решить:

$2R, r$

Найти:

T_1, T_2



$$2) \Delta U = 2mgR(1 - \cos\varphi) =$$

$$= \frac{2mgR\varphi^2}{2} = 2mg \cdot \sqrt{r^2 - e^2} \cdot \frac{\varphi^2}{2}$$

$$K = \frac{2m(r\dot{\varphi})^2}{2} = \frac{2mr^2}{2} \dot{\varphi}^2$$

$$\omega_2 = \sqrt{\frac{2mg\sqrt{r^2 - e^2}}{2mr^2}} = \sqrt{\frac{g\sqrt{r^2 - e^2}}{r^2}} \rightarrow T_2 = 2\pi \sqrt{\frac{r^2}{g\sqrt{r^2 - e^2}}}$$

$$1) \Delta U = \frac{2mgR\varphi^2}{2} = 2mg\sqrt{r^2 - e^2} \cdot \frac{\varphi^2}{2}$$

$$K = \frac{2mR^2\dot{\varphi}^2}{2} = \frac{2m\dot{\varphi}^2}{2} \cdot (r^2 - e^2)$$

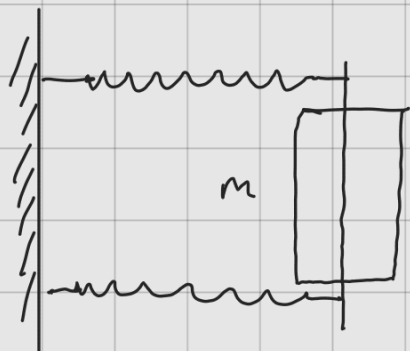
$$\omega_1 = \sqrt{\frac{2mg\sqrt{r^2 - e^2}}{2m(r^2 - e^2)}} = \sqrt{\frac{g}{\sqrt{r^2 - e^2}}} \rightarrow T_1 = 2\pi \sqrt{\frac{\sqrt{r^2 - e^2}}{g}}$$

N71

Дано:
 I, m, r
 κ

Найти: T

Решение:



$$\omega = \frac{\dot{x}}{R}$$

$$\dot{\omega} = \epsilon = \frac{a}{R} = \frac{\ddot{x}}{R}$$

$$\text{ЗСЭ: } \frac{2\kappa x^2}{2} + \frac{m\dot{x}^2}{2} + \frac{I}{2} \cdot \omega^2 = \text{const}$$

Продифференцируем по времени:

$$\frac{2\kappa x \cdot \dot{x} \cdot 2}{2} + \frac{2m\dot{x}\ddot{x}}{2} + \frac{I}{2} \cdot 2\dot{x} \cdot \frac{\ddot{x}}{R} = 0$$

$$2\kappa x + m\ddot{x} + \frac{I}{R^2} \ddot{x} = 0$$

$$\ddot{x} = -\frac{2\kappa}{m + \frac{I}{R^2}} x \Rightarrow \omega^2 = \frac{2\kappa}{m + \frac{I}{R^2}} \Rightarrow T = 2\pi \sqrt{\frac{m + \frac{I}{R^2}}{2\kappa}}$$

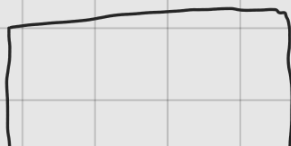
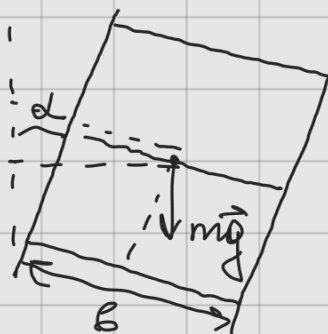
N72

Дано:

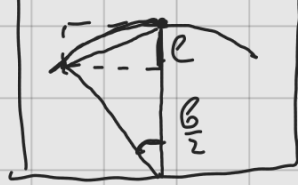
α, b

Найти: T

Решение:



= 1/2 m g sin alpha



→ m-b Körper

$$l = \frac{b}{2} (1 - \cos \varphi) ; \Delta h = \frac{b}{2} (1 - \cos \varphi) \sin \alpha$$

$$\Delta U = m g \frac{b}{2} (1 - \cos \varphi) \sin \alpha = \frac{m g b}{2} \sin \alpha \cdot \frac{\varphi^2}{2}$$

$$\Rightarrow K = \frac{I \omega^2}{2} = \left(\frac{m b^2}{12} + \frac{m b^2}{12} \right) \dot{\varphi}^2$$

$$\omega^2 = \frac{m g b}{2} \sin \alpha \cdot \frac{3}{m b^2} = \frac{3}{2} \frac{g \sin \alpha}{b}$$

$$\Rightarrow T = 2\pi \sqrt{\frac{2b}{3g \sin \alpha}}$$